MODEL COMPETITORS

PRESENTATION BY BRENDEN PARK

1. EnsembleBBB

Dataset Information (2023)

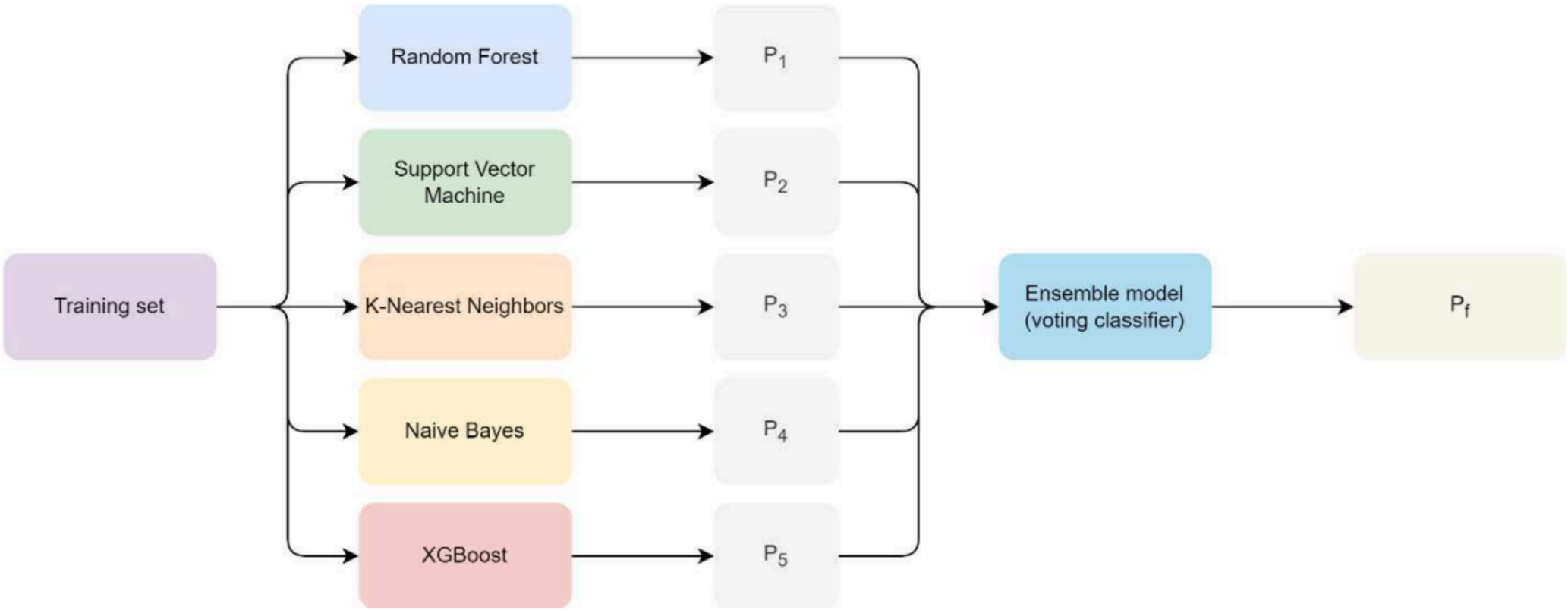
Dataset: 7807 compounds from the B3DB database was curated to remove redundant compounds, resulting in 7794 compounds (5455 BBB+/2339 BBB-)

Dataset Split: 70/30

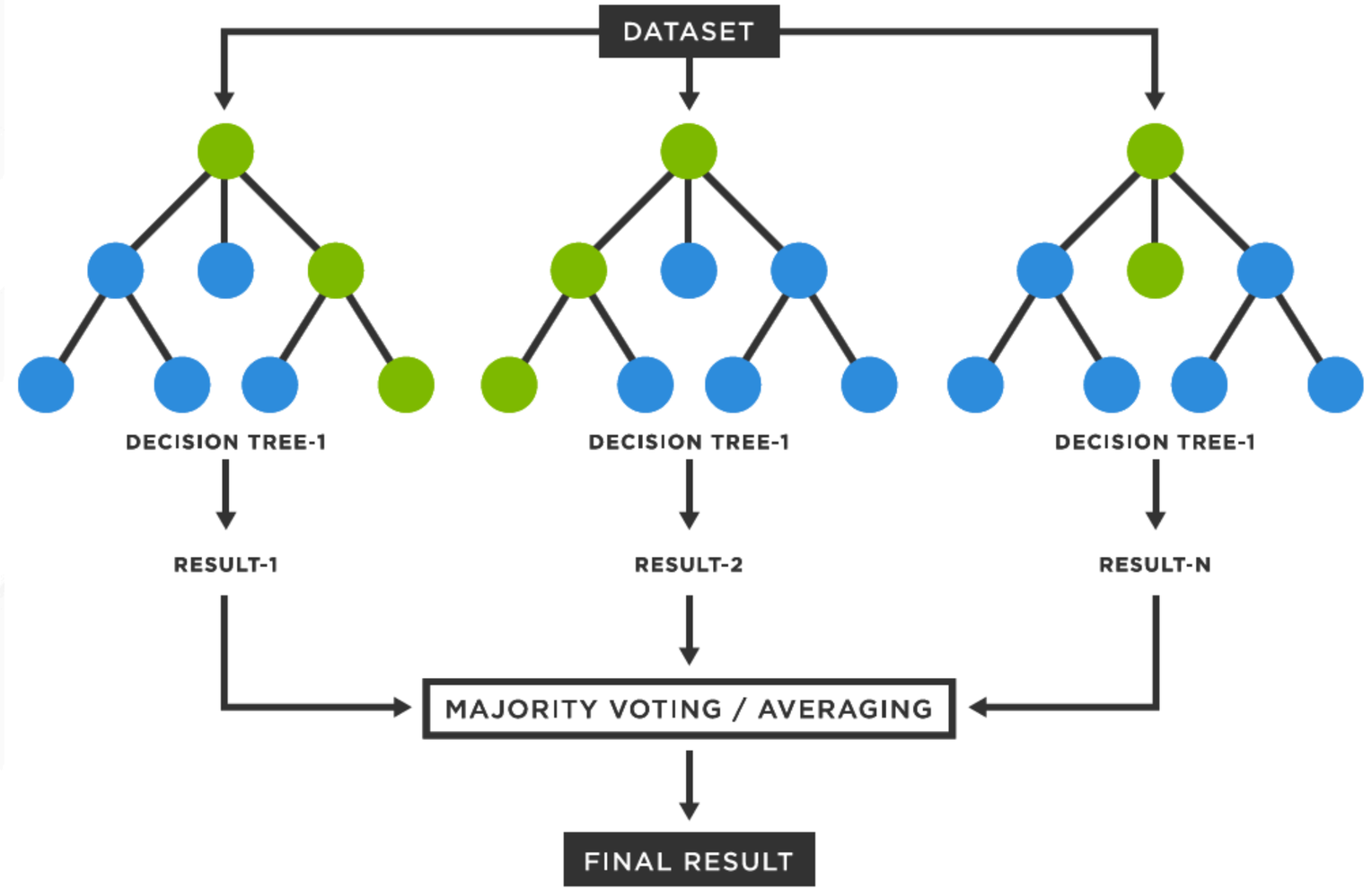
Dataset Encoding: MACCS (Molecular ACCess System keys) was used to convert chemical features to binary representations to be input alongside SMILES strings

1. EnsembleBBB

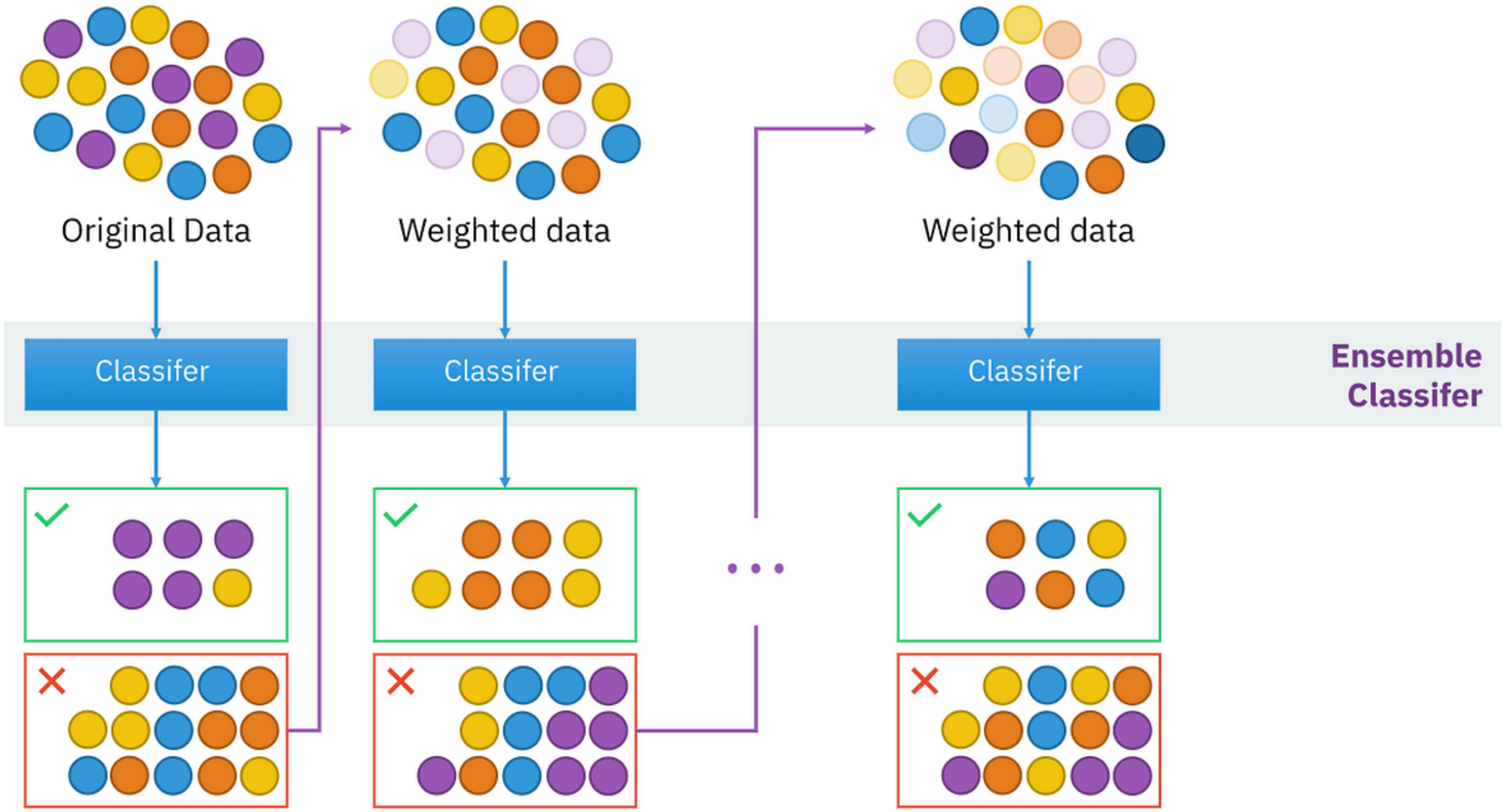
Model Architecture



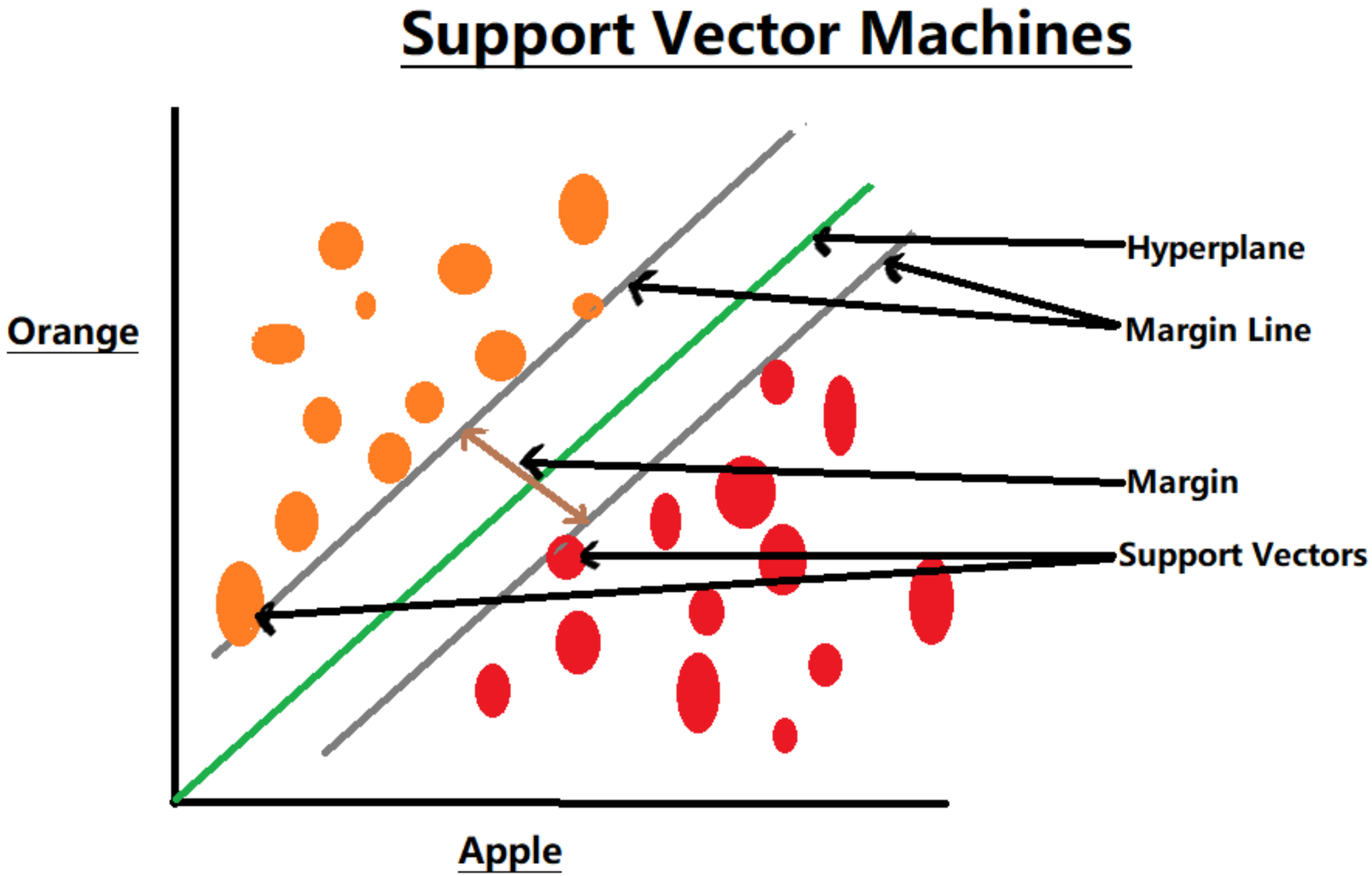
Random Forest



XGBoost


Original Data
Weighted data
Weighted data
Classifer
Classifer
Classifer
Ensemble Classifer

Support Vector Machines



1. EnsembleBBB

Model Results

Table 3. Statistical performance metrics to evaluate each classification model.

Dataset	Ensemble model	Fingerprint	SE	SP	ACC	F1 score	MCC	Mean AUC 5-fold CV
Training set	Ensemble_1	Substructure	0.97	0.98	0.97	0.97	0.95	0.96
	Ensemble_2	MACCS	0.98	0.98	0.98	0.98	0.96	0.97
	Ensemble_3	PubChem	0.98	0.99	0.99	0.99	0.97	0.97
	Ensemble_4	AtomPairs2D	0.95	0.94	0.94	0.94	0.88	0.95
	Ensemble_5	CDKextended	0.98	0.99	0.99	0.99	0.98	0.96
	Ensemble_6	EState	0.92	0.90	0.91	0.91	0.82	0.94
Testing set	Ensemble_1	Substructure	0.92	0.80	0.88	0.91	0.74	0.92
	Ensemble_2	MACCS	0.93	0.82	0.89	0.91	0.76	0.94
	Ensemble_3	PubChem	0.93	0.82	0.89	0.91	0.75	0.92
	Ensemble_4	AtomPairs2D	0.90	0.76	0.85	0.88	0.66	0.89
	Ensemble_5	CDKextended	0.91	0.80	0.87	0.90	0.72	0.92
	Ensemble_6	EState	0.89	0.75	0.84	0.88	0.65	0.89

2. DeePred BBB

Dataset Information (2022)

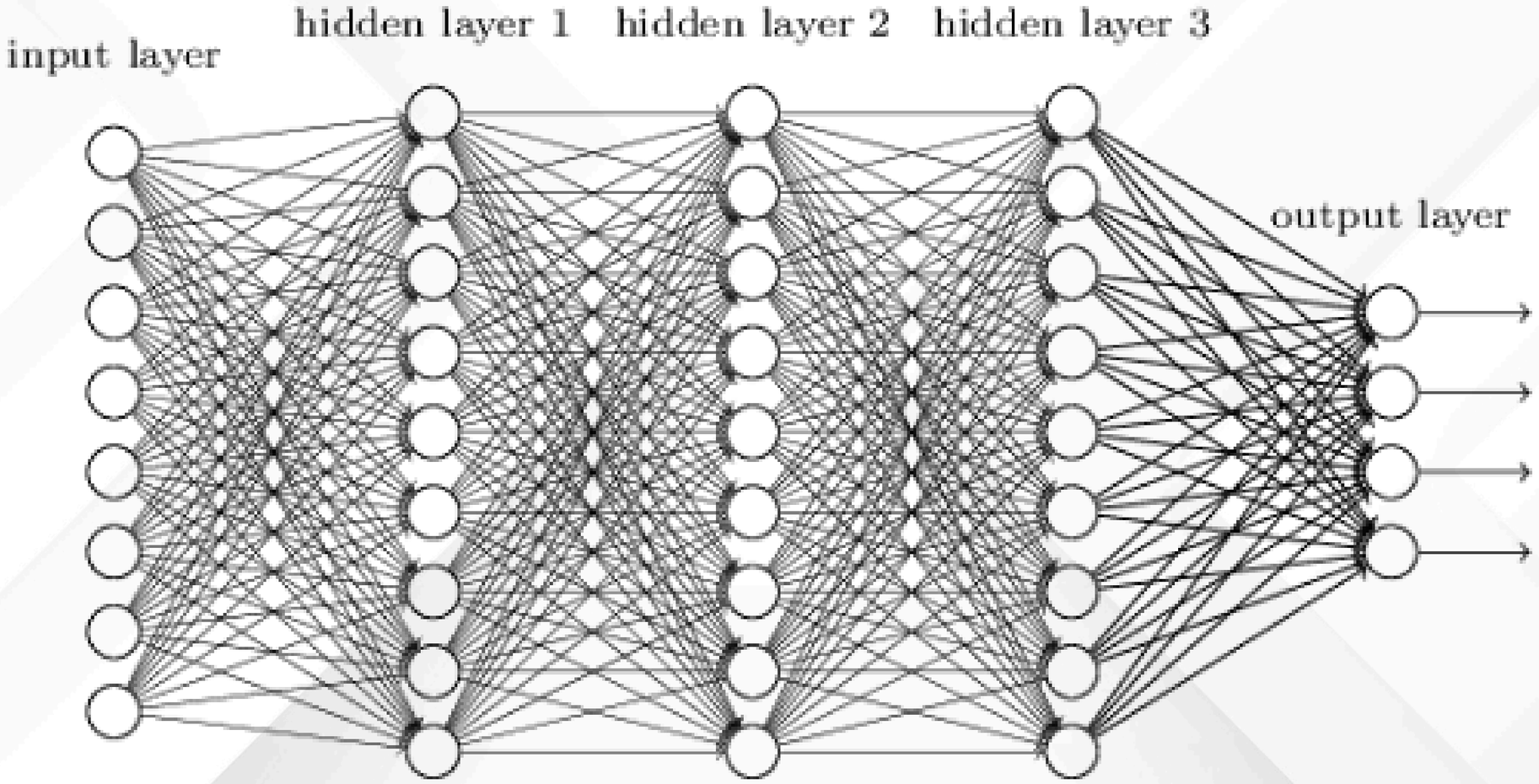
Dataset: 3971 compounds from 3 open-access databases was curated to 3605 non-redundant compounds (2607 BBB+/998 BBB-)

Dataset Split: 75/15

Feature Calculation: Physiochemical properties (physical and chemical information), MACCS fingerprints (atomic structure information), and substructure fingerprints (functional groups) were calculated for each compound

2. DeePred BBB

Model Architecture



2. DeePred BBB

Hyperparameter Selection

Parameter	Values
Number of hidden layers	1–5
Number of neurons	100, 200, 300, 500, 800
Dropout ratio	0.1, 0.2, 0.3, 0.4, 0.5
Learning rate	0.0001, 0.0002, 0.0003, 0.001, 0.002, 0.003
Epochs	100, 200, 400, 500, 800

2. DeePred BBB

Results

Algorithm	AUC	AUPRC	AP	F1	A (%)	HD
SVM (Polynomial $d = 3$)	0.948	0.965	0.965	0.98	96.01	0.029
SVM (Sigmoid)	0.921	0.971	0.944	0.962	94.45	0.055
SVM (Linear)	0.916	0.969	0.938	0.963	94.56	0.054
NB	0.844	0.948	0.899	0.935	90.18	0.098
kNN (3)	0.927	0.974	0.949	0.968	95.3	0.047
RF (3, 20)	0.815	0.943	0.887	0.938	90.29	0.0971
DNN	**0.992**	**0.997**	**0.996**	**0.987**	**98.07**	**0.019**

3. Dee-B^3

Dataset Information (2022)

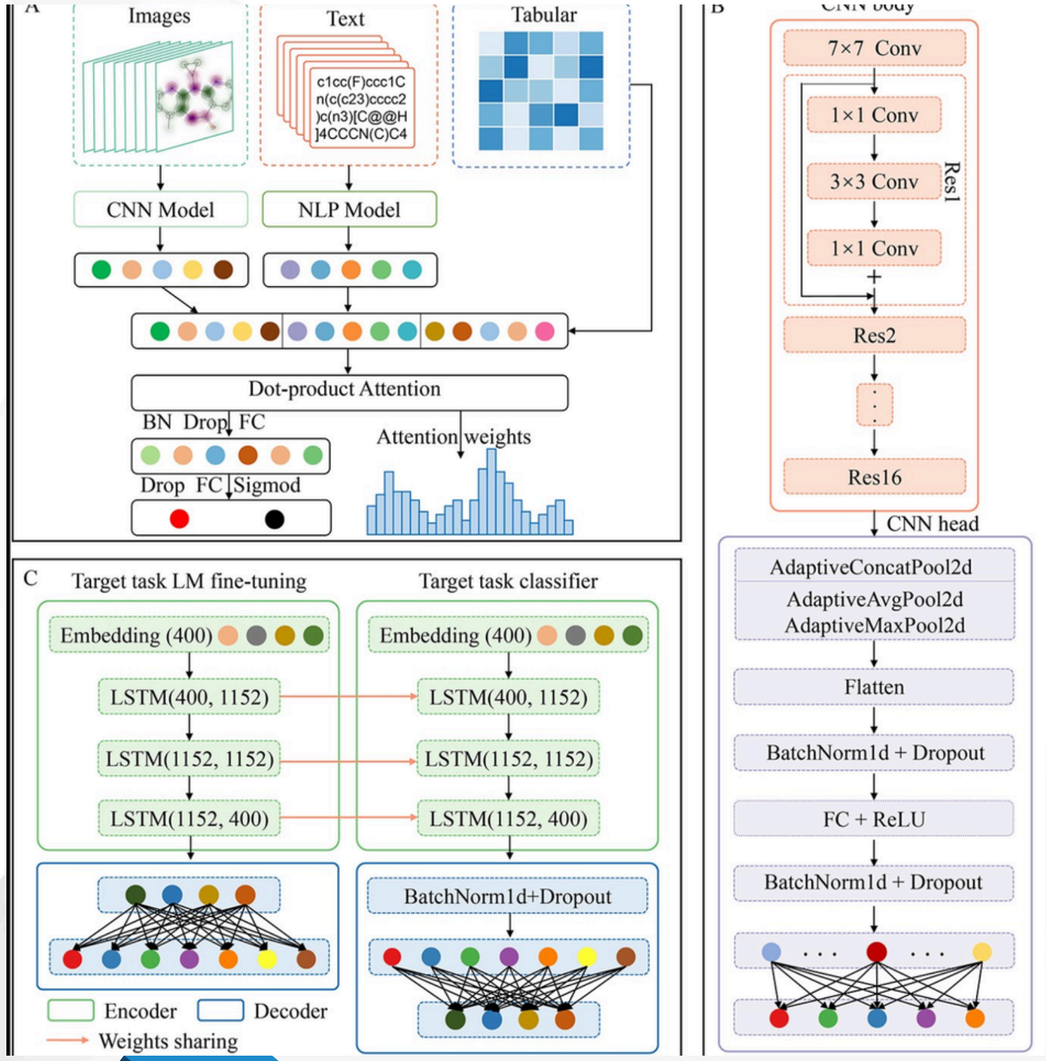
Training Dataset: 4364 non-redundant, organic compounds (3125 BBB+/1239 BBB-) was organized from 6 literature sources

Testing Dataset: 2670 non-redundant, organic compounds (1258 BBB+/1412 BBB-) was organized from 4 literature sources

Feature Calculation: The canonical SMILES string, molecular graph, and 1D/2D molecular descriptors were calculated for each compound in the dataset

3. Dee-B^3

Model Architecture



3. Dee-B^3

Results

Model	SN (%)	SP (%)	MCC	ACC	AUC
lightBBB	86	55	0.43	0.70	–
Deep-B^3	85	64	0.49	0.74	0.83

4. B3clf

Dataset Information (2025)

Training Dataset: 7807 compounds from the B3DB database was curated to remove compounds with invalid 3D representation, resulting in 7407 compounds (4855 BBB+/2552 BBB-)

Testing Dataset: CNS drugs from DrugBank, H1-antihistamines (allergy medication), and discontinued compounds were curated to create an external dataset of 171 BBB+ and 4 BBB-

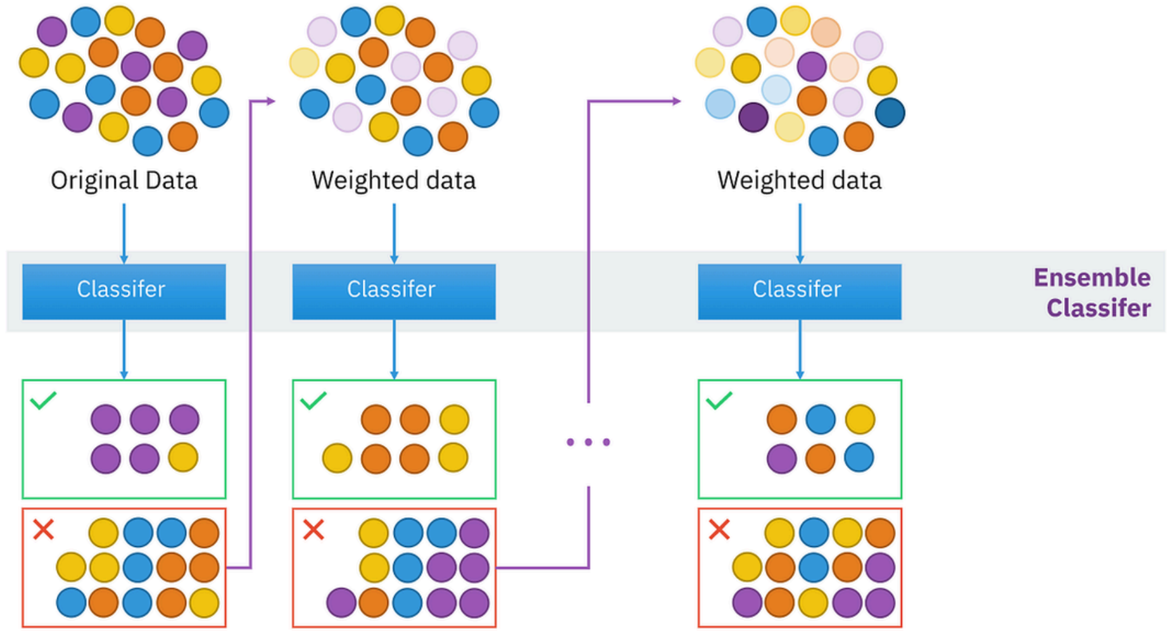
4. B3clf

Dataset Information (2025)

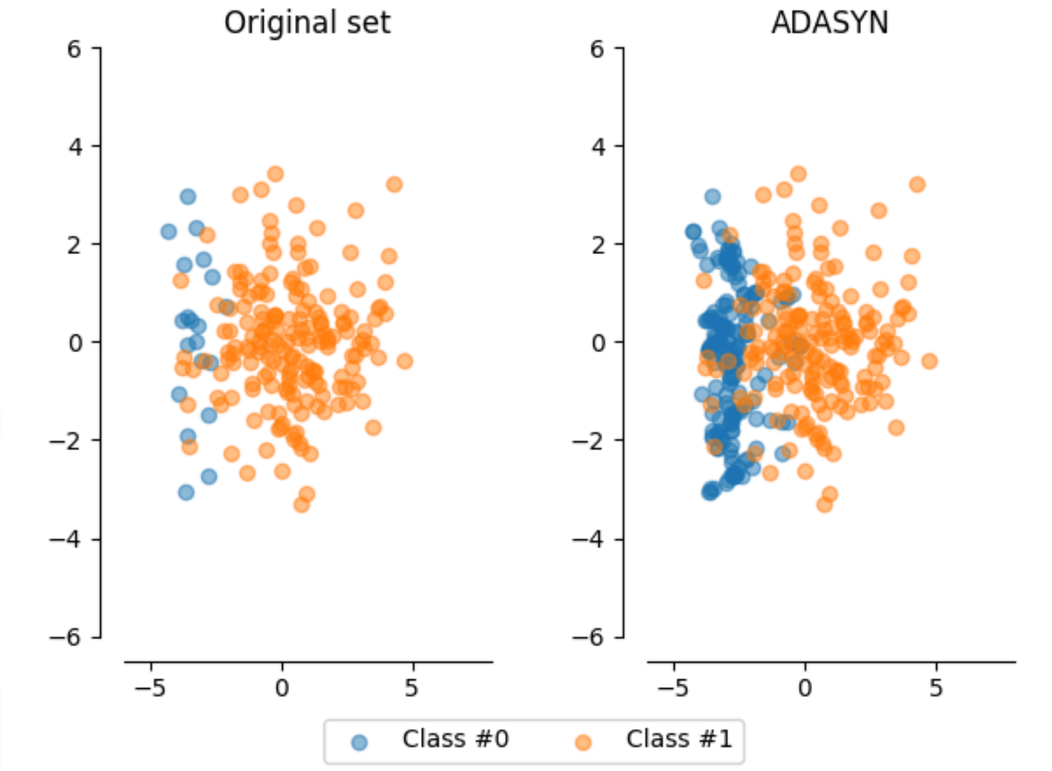
Molecular Feature: 1D, 2D, 3D non-redundant descriptors were calculated for each chemical compound

4. B3clf

Model Architecture



XGBoost



ADASYN (Adaptive Synthetic Sampling)

4. B3clf

Model Results

Source	Model	AP	Accuracy	Sensitivity	Specificity	Precision	F1	MCC	GEOM	BACC	ROC_AUC
Present study	dtree-borderline_SMOTE	0.9987	0.8114	0.8070	1.0000	1.0000	0.8932	0.2954	0.8983	0.9035	0.9934
Present study	xgb-classic_ADASYN	0.9993	0.9086	0.9064	1.0000	1.0000	0.9509	0.4258	0.9521	0.9532	0.9708
Present study	xgb-borderline_SMOTE	0.9991	0.8457	0.8421	1.0000	1.0000	0.9143	0.3296	0.9177	0.9211	0.9605
Present study	xgb-classic_RandUndersampling	0.9991	0.7943	0.7895	1.0000	1.0000	0.8824	0.2810	0.8885	0.8947	0.9635
Present study	knn-classic_RandUndersampling	0.9988	0.8743	0.8772	0.7500	0.9934	0.9317	0.2725	0.8111	0.8136	0.9518
Present study	xgb-classic_SMOTE	0.9988	0.8629	0.8596	1.0000	1.0000	0.9245	0.3504	0.9272	0.9298	0.9488
Swanson et al. (2024)[78]	GNN	0.9988	0.9200	0.9298	0.5000	0.9876	0.9578	0.2368	0.6818	0.7149	0.9518
Present study	knn-classic_SMOTE	0.9986	0.8343	0.8304	1.0000	1.0000	0.9073	0.3173	0.9113	0.9152	0.9444
Present study	knn-common	0.9986	0.9257	0.9357	0.5000	0.9877	0.9610	0.2483	0.6840	0.7178	0.9444
Present study	xgb-common	0.9985	0.8743	0.8713	1.0000	1.0000	0.9313	0.3661	0.9335	0.9357	0.9386
Present study	dtree-common	0.9982	0.8743	0.8772	0.7500	0.9934	0.9317	0.2725	0.8111	0.8136	0.9313
Present study	logreg-classic_RandUndersampling	0.9981	0.8857	0.8889	0.7500	0.9935	0.9383	0.2880	0.8165	0.8194	0.9225
Present study	dtree-kmeans_SMOTE	0.9980	0.8686	0.8713	0.7500	0.9933	0.9283	0.2654	0.8084	0.8107	0.9247
Present study	knn-kmeans_SMOTE	0.9980	0.8914	0.9006	0.5000	0.9872	0.9419	0.1924	0.6710	0.7003	0.9167
Present study	knn-borderline_SMOTE	0.9979	0.7200	0.7135	1.0000	1.0000	0.8328	0.2320	0.8447	0.8567	0.9152
Present study	logreg-classic_ADASYN	0.9979	0.8343	0.8304	1.0000	1.0000	0.9073	0.3173	0.9113	0.9152	0.9137
Present study	xgb-kmeans_SMOTE	0.9979	0.8629	0.8596	1.0000	1.0000	0.9245	0.3504	0.9272	0.9298	0.9137
Present study	knn-classic_ADASYN	0.9977	0.7029	0.6959	1.0000	1.0000	0.8207	0.2230	0.8342	0.8480	0.9079
Present study	logreg-borderline_SMOTE	0.9977	0.8457	0.8421	1.0000	1.0000	0.9143	0.3296	0.9177	0.9211	0.9050
Present study	logreg-common	0.9977	0.8914	0.9006	0.5000	0.9872	0.9419	0.1924	0.6710	0.7003	0.9064
Present study	logreg-kmeans_SMOTE	0.9975	0.8857	0.8947	0.5000	0.9871	0.9387	0.1854	0.6689	0.6974	0.8977
Present study	logreg-classic_SMOTE	0.9968	0.8571	0.8713	0.2500	0.9803	0.9226	0.0537	0.4667	0.5607	0.8713
Parakkal et al. (2022)[79]	Mixed DL (10-fold CV)	0.9964	0.8514	0.8538	0.7500	0.9932	0.9182	0.2461	0.8002	0.8019	0.8757
Parakkal et al. (2022)[79]	Mixed DL	0.9958	0.9200	0.9240	0.7500	0.9937	0.9576	0.3495	0.8325	0.8370	0.8582
Kumar et al. (2022)[80]	DNN	0.9957	0.7829	0.7836	0.7500	0.9926	0.8758	0.1899	0.7666	0.7668	0.8567
Han et al. (2025)[81]	AutoML-HSL	0.9923	0.9200	0.9298	0.5000	0.9876	0.9578	0.2368	0.6818	0.7149	0.8180
Present study	dtree-classic_ADASYN	0.9922	0.6800	0.6784	0.7500	0.9915	0.8056	0.1360	0.7133	0.7142	0.7412
Present study	dtree-classic_RandUndersampling	0.9841	0.7771	0.7778	0.7500	0.9925	0.8721	0.1862	0.7638	0.7639	0.7186
Present study	dtree-classic_SMOTE	0.9820	0.7371	0.7544	0.0000	0.9699	0.8487	-0.0859	0.0000	0.3772	0.4620
Tang et al. (2022)[82]	Multi-model		0.8343	0.8343		1.0000	0.9097	0.0000			

THANK YOU